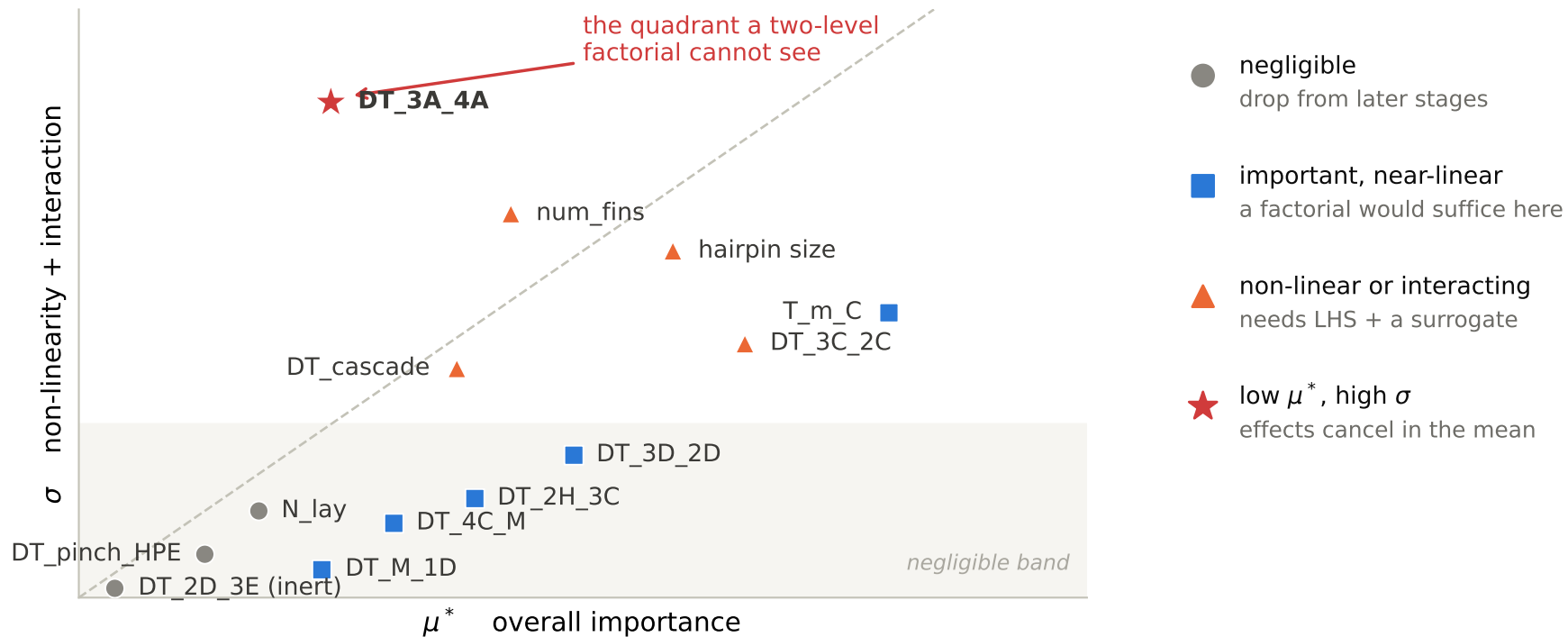


Morris screening — plan, not results

Placements are HYPOTHESES for the run to test. They are falsifiable, which is half the reason to do it.



Reading the plane

μ^*

mean ABSOLUTE elementary effect — how much the output moves when this factor moves, anywhere in the space. A ranking of importance.

σ

standard deviation of those effects. Large σ means the effect DEPENDS on where you are: either the response curves, or another factor is changing it. Morris cannot separate the two.

diagonal

$\sigma = \mu^*$. Above it a factor varies by more than its own average size — the conventional threshold for "a linear model of this factor is not safe".

ΔT_{3A4A}

ψ has an interior optimum near 20 K, so its elementary effects are positive below and negative above and the SIGNED mean nearly cancels. A two-level factorial computes that signed mean and calls the factor unimportant. It is the one factor we already know has an optimum.

Which factors are expected to drive which KPI

Qualitative, and the point of the screen is to correct it. Note that no two columns have the same ranking.

	η_{RTE}	ψ	wells per MWe	UA per MWe	ρ_p kW/m ³	kW per well	ϵ_{RTE}	ΔE_{el} per well	deviation
hairpin size					▲	▲			▲
num_fins			▲		▲	▲	▲		▲
T_m_C									
DT_3C_2C							▲		
DT_3D_2D									
DT_cascade							▲		▲
N_lay							▲		
DT_4C_M									
DT_M_1D									
DT_pinch_ORC				▲					
DT_2H_3C									
DT_3A_4A		▲	▲						
t_ch							▲		▲

none
weak
moderate
strong
▲ expected non-linear or interacting

Read the columns, not the rows. ΔT_{3A4A} is strong for η_{RTE} and for the well count and almost absent from UA;
 $\Delta T_{pinch, ORC}$ is the reverse. A study that ranks designs on one column will recommend the corner that is free in that
column and expensive in the others — which is how η_{RTE} alone buys efficiency with exchanger area, and COP with boreholes.

KPI register for the screen

Eight of these do not exist in v0.15 and are cheap to add. Four of the additions are regime labels, not performance numbers.

OBJECTIVES — the four that must always be read together

eta_RTE	round trip, both parasitics charged	have
psi	second law, whole geothermal resource charged	have
wells_per_MWe	(N_wells + N_geo) per MWe — the borehole cost	have
UA_per_MWe	total conductance per MWe — the exchanger cost	have

SECOND OBJECTIVE TIER — density and productivity

rho_P_kW_m3	electric power per m3 of store	have
rho_P_kW_per_m	electric power per metre of borehole	have
kW_per_well	electric power per well = W_{el_out} / N_{wells}	ADD
dE_elec_kWh	net electric energy per well per cycle	have
rho_E_kWh_m3	thermal energy density	have
eps_RTE	eta_RTE x the fraction of the store that cycles	have
I_per_MWh	exergy destroyed per MWh delivered — one scalar for the audit	ADD

CONSTRAINTS — decide whether a run is COMPARABLE, not whether it is good

deviation	delivered vs target; a run far off is a different plant	ADD to KPI dict
pinch_hpe/hpc/orce/orcc	which exchanger binds, and whether any crossed	have
T_3e_binding	which rule set the boiling temperature — a REGIME LABEL	ADD
merge_proximity_max	H3 saturation; differs across runs, breaks comparability	ADD to KPI dict
eps_cycled	is the store being used at all	have
feasible	validate_case returned no error	ADD

DIAGNOSTICS — explain the main effects, do not rank on them

I_<component>	destruction by component, 20 entries	have
resource_utilisation	fraction of the geothermal exergy actually used	have
N_geo, m_geo_kg_s, T_reinject_C	the source side	have
f_pump	parasitic fraction of gross output	have
exergy_gates_pass	the audit closed	have

The four objectives are not combinable without cost weights. Report the Pareto front over them, or state the weights explicitly.
The constraint block is what stops the screen comparing plants that are not the same plant: a run whose deviation is +30 %, or whose ORC switched from pinch-bound to hot-end-bound, is not a worse design — it is a different question.